

Rafael Polo

Senior Data & Platform Engineer · Tech Lead · 20+ years

Allschwilerstrasse 77, 4055 Basel · +41 77 423 59 48 · rafael@extrapolo.com linkedin.com/in/rafaelpolo · github.com/rafapolo · extrapolo.com Swiss B permit · EU citizen (Portuguese / Brazilian)

PROFILE

Senior software engineer with 20+ years delivering data infrastructure, full-stack platforms, and AI-powered tools. Swiss work experience at **Swisscom** (Zürich) and **EDGE Empower** (Zug). I own problems end-to-end — from data-pipeline design and system architecture to production deployment and stakeholder communication — and I am used to high-availability, regulated environments.

Core stack: Python · TypeScript · React · Node.js · Ruby on Rails · PostgreSQL · Docker · Kubernetes · AWS · CI/CD · LLM · RAG · LangChain

PROFESSIONAL EXPERIENCE

DevOps Engineer — Swisscom

03/2024 – 03/2025 · Zürich, Switzerland

Switzerland's largest telecom operator; the platforms I maintained served field technicians across the country.

- Maintained and modernized **28 mission-critical internal services** used daily by field-operations teams
- Delivered new features in Ruby on Rails within a SAFe Agile environment
- Deployed and operated services with Docker and Kubernetes under strict SLAs where production reliability was non-negotiable
- Contributed to CI/CD improvements supporting frequent, low-risk releases

Senior Technical Lead — EDGE Empower® (EDGE Strategy)

02/2023 – 02/2024 · Zug, Switzerland

Global certification body measuring inclusion and gender equality at work; **first in-house technical hire** in a team of thirty.

- Built a modern offline PayTool replacing the legacy system, **cutting processing time by 98%**
- Parallelized the CI/CD pipeline, **reducing build time from 14 to 4 minutes**
- Contracted and coached 3 external developers and a UX designer to accelerate delivery
- Translated a years-long outsourced backlog into an owned, in-house product roadmap
- Managed servers, databases, and critical security upgrades across the infrastructure

Independent Consulting — ExtraPolo

01/2008 – 01/2023 · Brazil · Germany · Greece · Portugal

40+ projects for NGOs, journalists, public agencies, and cultural institutions — data engineering, investigative platforms, and web systems, including engagements with The Intercept Brasil, Tactical Technology Collective (Berlin), and the Rio de Janeiro Environmental Secretariat.

Earlier Roles (2003 – 2014)

Web Developer, trommsdorff + drüner — Berlin, 2013–2014 — Ruby/TypeScript features, SQL optimization

Software Engineer, Cortex Intelligence — Rio de Janeiro, 2007–2008 — SaaS features, ETL crawlers, Java **Co-founder**, MeuServidor — 2003–2004 — Linux/PHP hosting company; acquired

SELECTED PROJECTS

- **Rodado** — pipeline indexing ~730 official Brazilian public datasets (1TB Parquet), queryable via DuckDB and natural-language-to-SQL over MCP. github.com/rafapolo/rodado
- **HelvetiScan** — scanner and mapper of 2.5M+ Swiss .ch domains (HTTP, DNS, TLS, WHOIS, CVE); sector classification and risk benchmarking. github.com/rafapolo/helvetiscan

40+ further open-source projects — browse the interactive graph at extrapolo.com · github.com/rafapolo

TECHNICAL SKILLS

Area

Languages

Data

AI / LLM

Frontend

DevOps

Backend & APIs

Tools & Technologies

Python · Ruby · TypeScript · JavaScript · SQL · Rust

ETL pipelines · PostgreSQL · DuckDB · BigQuery · Parquet · data modeling

AI agents · RAG · LangChain · Ollama · natural-language SQL interfaces

React · D3.js · Deck.gl · MapLibre

Docker · Kubernetes · CI/CD · AWS · Linux · Git · SAFe Agile

Ruby on Rails · Node.js · Bun · REST architecture

EDUCATION

Postgraduate Degree, Data Science for Finance — NOVA IMS, Lisbon · 2021 **MSc, Information Systems** (*Knowledge Representation & Reasoning*) — PPGI/UNIRIO, Rio de Janeiro · 2010–2012 **BSc, Computer Science** (*Thesis: Agile Systems Development for Web 2.0*) — Universidade Metodista Bennett, Rio de Janeiro · 2004–2009

CERTIFICATIONS

Swiss Federal Act on Data Protection (nFADP) · Databricks for Data Engineering · Generative AI & LLMs: Architecture and Data Preparation · Gen AI Foundational Models for NLP · AI Agents Using RAG and LangChain

LANGUAGES

German

B1 (telc)

English

Full professional

Spanish

Full professional

Portuguese

Native

Greek

A2

ABOUT

Based in Basel since 2023; I have lived and worked in five countries — Rio de Janeiro, Berlin, Athens, Lisbon, and Zürich — and adapt quickly to new teams and environments.

Volunteer: Three years teaching computer science to refugees in Athens (2017); helped build a computer lab.

Personal: Passionate cyclist — once rode from Berlin to Athens.